concentration and assay, the uranium is ready for fuel refabrication. The partially decontaminated thorium is concentrated and stored.

(ii) The separated silicon-carbide coated fissile particles from the head end process for HTGR fuel are mechanically crushed to expose the fuel and are burned to remove carbon and oxidize the fuel material: the ash is leached to separate the fuel and fission products from the coating waste. The ^{235}U is separated from the fission products by solvent extraction using a PUREX flowsheet. An organic solvent containing only 3–5% TBP is used in order to avoid criticality problems. A reductant such as Fe(II) sulfamate is added to the feed in order to force the small amount of plutonium present into the aqueous raffinate stream. The waste streams are treated as in the PUREX process.

The process is plagued by both chemical and nuclear difficulties. The decay chain $^{233}Th \rightarrow {}^{233}U$ forms 27 d half-life ^{233}Pa. For a complete decay of all ^{233}Pa to ^{233}U, the spent fuel elements must be cooled for about a year. A still considerable amount of longlived ^{231}Pa ($t_{½}$ 32 760 y) is present in the spent fuel (about 1/2000 of the amount of ^{233}U); protactinium complicates the reprocessing chemistry and constitutes an important long-lived waste hazard.

The ^{233}U isolated contains some ^{232}U formed through reactions indicated in Figure 20.3. Since the half-life of ^{232}U is rather short and its decay products even shorter lived, a considerable γ-activity will grow in with the ^{233}U stock, complicating its handling. Since the first decay product is ^{228}Th, some ^{228}Th forms in the fuel elements, making the ^{232}Th contaminated by this isotope. Thus neither the ^{233}U nor the ^{232}Th produced can be free from γ-activity.

21.8. WASTES STREAMS FROM REPROCESSING

Fuel reprocessing generates a large variety of wastes which can be classified in categories according to activity (low, medium and high), physical state (gas, liquid or solid package), or decay characteristics (shortlived, longlived), each treated separately. The amounts of different categories are given in Table 21.8; these wastes are treated internally at the plant and not released to the environment. Because reprocessing plants vary considerably the amounts of wastes produced also differs, especially for the liquid medium and low level categories. In this section we briefly discuss the various waste streams appearing at the plant and the treatment methods.

21.8.1. Gaseous wastes

Gaseous wastes originate mainly from the chopping and dissolution operations. In current practice the volatile radionuclides are discharged to the stack after scrubbing with sodium hydroxide and filtration through a special zeolite or charcoal filter. The hydroxide scrubbing removes the acidic nitrous oxides which pass through the

Table 21.8 Typical annual amounts of waste arising in a reprocessing plant (per t IHM and 3 years cooling time)

Type	Form	Volume (m^3)	β,γ-activity (TBq)	Pu-content (kg)
HLLW	liquid	5	10^5	~0.05
MLLW	liquid	5		
LLLW	liquid	100		
HLLW	conc. liquid	0.5	10^5	0.05
Fission products	glass	0.15	10^5	0.05
Hulls	concrete	0.8	400	0.01
Ion/exch., precip.	Bitumen	0.8	~100	0.008
Iodine	concrete	0.03	<0.01	
ML and LL α	solid	0.1		
Technical waste	concrete	3.2	~0.5	~0.01

recombination unit above the dissolver. The silver impregnated zeolite or charcoal removes remaining traces of iodine.

The most hazardous volatile constituents are the iodine and ruthenium fission products. Though more than 95% of the iodine is volatilized in the dissolver (as I_2, HI and HIO mainly) most of it is caught in the off-gas scrubber and most of what remains is removed by the filters. With these techniques the retention of iodine in the plant is >99.5%.

Ruthenium forms volatile RuO_4 in the dissolver. Almost all RuO_4 is retained in the gas purification system. As an additional feature some plants use a steel wool filter (after the nitric acid recombination) to catch the RuO_4.

Of the noble gases, radioactive xenon has completely decayed after 1 y cooling, but krypton contains ^{85}Kr with 10.7 y half-life. This isotope is produced in appreciable amounts, and though commonly it has been released to the atmosphere, this is no longer acceptable. Many processes have been devised for krypton removal. Krypton in dry, clean air is effectively trapped on a charcoal filter at cryogenic temperature; however, because of explosion risk (due to reaction between radiolytically formed ozone and carbon), the favored process is condensation by liquid N_2 (krypton boils at −153°C) followed by fractional distillation. This removes >99% of ^{85}Kr. The krypton can be stored in pressurized cylinders until ^{85}Kr has decayed (>100 years).

The amount of tritium released in reprocessing is considerable. In the chopping section it is released as T_2, but as HTO in the dissolver, where >90% of all tritium formed is present. While T_2 can be caught (particularly if chopping is done in air), a recovery of the HTO formed in the dissolver would be expensive. Where it cannot be released to the environment the tritium can be trapped. In one process, called Voloxidation, the chopped fuel elements are treated with oxygen at 450−700°C before dissolution. Tritiated water is generated which should be relatively free of ordinary water and consequently occupy a much smaller volume than tritiated wastes do in present

plants. Voloxidation may collect ~99% of the tritium present in unprocessed fuel. Even when all tritium is released to the environment, some precautions against its spreading in the plant are usually necessary in order to reduce the dose to operators. By proper design of the first extraction cycle most of the tritium can be confined to this part of the plant.

^{14}C is formed through the $^{14}N(n,p)^{14}C$ reaction in the nitrogen contained in the fuel elements. It is released mainly as CO_2 at the dissolution. Though only a small amount is formed (~40 GBq/GW_ey for each ppm nitrogen in the fuel), its release to the environment, makes it the dominating dose commitment of the fuel cycle back end (from reprocessing in 1989, 97 manSv from ^{14}C, and 98 manSv from all other nuclides). Presently $^{14}CO_2$ is released to the atmosphere, but techniques for its retention are available. It will probably finally be caught as $CaCO_3$, with a retention of ~80%. However, the future use of nitride fuels will increase this proportional to the ^{14}N content (as mentioned above, ~40 GBq/GW_ey for each ppm of ^{14}N in the nitride fuel, i.e. ≈2 PBq/ GW_ey for UN made from natural nitrogen).

21.8.2. Liquid wastes

The high level liquid PUREX waste (HLLW) contains typically >99.5% of the FPs, <0.5% of the U and <0.2% of the Pu present in the fuel as ~1 M HNO_3 solution, see Table 21.9. It is pumped to storage tanks as discussed further in §21.10. Total actual Pu losses during reprocessing is presently ≤0.2% of the feed.

The medium level liquid waste (MLLW) results essentially from evaporating various streams from the chemical process, such as solvent clean-up, off-gas scrubbers, product concentration, etc; Table 21.8. It may contain up to 0.5% of the uranium and up to 0.2% of the plutonium processed. The radioactivity is usually <40 GBq/l (an average value is 4 GBq/l). The solutions also contain appreciable amounts of solids (e.g. $NaNO_3$, iron, etc.). The waste may be neutralized and is stored in steel tanks at the reprocessing site.

Liquid waste is generated in numerous places with activities <0.1 GBq/m^3. Such waste is classified as low level. Some of these liquids may be clean enough to be released directly into the environment. Others are cleaned by flocculation, ion exchange, sorption, and similar processes. The general philosophy for liquid wastes is to concentrate all radioactivity to the next higher level because the waste volumes decrease in the order LLLW > MLLW > HLLW. Thus, in principle, the three kinds of wastes are reduced to two (HLLW and MLLW) and cleaned aqueous effluent. The MLLW and residues from LLLW cleaning are treated as the wastes of the nuclear power stations, i.e. concentrated and put into a disposal matrix such as concrete or bitumen (see §20.4.3). At some coastal sites it has been the practice to release the LLLW to the sea, with official permission. The nuclides of main concern are ^{3}H, ^{90}Sr, ^{137}Cs, ^{106}Ru, and the actinides.

Table 21.9 Composition of HLLW waste from PUREX reprocessing of 1 t IHM LWR fuel with a burnup of 33 000 $MW_{th}d/t$ IHM and a cooling time of 3 years

Component		Weight (kg) in original waste volume (~5 m^3)	Approx. molarity in 0.5 m^3 concentrate
H^+		1.4	~1.0
NO_3^-		900	~2.4
Fission products:	Group I (Rb, Cs)	2.94	0.046
	Group II (Sr, Ba)	2.37	0.041
	Group III (Y, Ln)	10.31	0.15
	Zr	3.54	0.076
	Mo	3.32	0.068
	Tc	0.77	0.016
	Group VIII (Ru, Rh, Pd)	4.02	0.078
	Te	0.48	0.0075
	Others	0.35	0.004
	Total fission products	28.1	0.487
Corrosion products:	Fe	1.1	0.04
	Cr	0.2	0.008
	Ni	0.1	0.003
	Total corrosion products	1.4	0.051
Phosphate (from TBP)		0.9	0.02
Actinides:	U (~0.5%)	4.8	0.040
	Np (~100%)	~0.44	0.0037
	Pu (~0.2%)	~0.018	0.00015
	Am (100%)	~0.28	0.0023
	Cm (100%)	~0.017	0.00014
	Total actinides	5.5	0.047
Neutron poison (e.g. Gd)		12	0.15

21.8.3. Organic wastes

The liquid organic waste consists of spent TBP diluent mixtures originating from the organic solvent clean-up circuits and from the diluent (kerosene) washings of the aqueous streams (to remove entrained solvent); in addition to degradation products of TBP and the diluent it contains small amounts of actinides (mainly U and Pu) and FP (mainly Ru, Zr, and Nb).

This type of waste is disposed of by incineration or decomposed by hydrolysis and pyrolysis leading to the formation of inactive hydrocarbons, which are distilled, and active phosphoric acid, which is treated together with other aqueous wastes.

21.8.4. Solid wastes

High level solid waste (HLSW) originates at the dissolver. The hulls from the dissolution contain activation products and small amounts of undissolved fuel ($\leq$0.1%). The

dissolver solution contains finely divided particles of undissolved seminoble metal alloys (Ru, Rh, Mo, Pd, etc.). This suspension is treated by filtering or centrifugation prior to the solvent extraction. Past practice in the USA and the former USSR was to put the HLSW in shielded containers which are transported to and stored at a dry disposal site. In the future the same disposal is expected to be used for the HLSW as for the solidified HLLW (§21.12).

Medium and low level solid wastes are produced at numerous places. They are divided in various ways: combustible, noncombustible, α-bearing, non-α-bearing, etc., and treated independently, when possible, to reduce volume. The wastes are then fixed in the disposal matrix (§20.4.3). Table 21.8 gives the relative amounts of solid waste produced in reprocessing. The final deposition of these wastes is further discussed in §21.13.

21.8.5. Environmental releases from reprocessing plants

As described in §21.8.1 relatively large amounts of radioactive gases with a low hazard index (^{3}H, ^{14}C and ^{85}Kr) are released to the environment, see Table 21.10. Other large

Table 21.10 Annual releases to the environment from some reprocessing plants

Source		Half-life (y)	Sellafield† (TBq)	PUREX‡ (TBq/GW_ey)
^{3}H	Airborne	12.33	222	41
	Liquid		1050	643
^{14}C	Airborne	5730	4.1	2.0
	Liquid			0.54
^{85}Kr	Airborne	10.72	26000	12300
^{90}Sr		28.5	600	11
^{95}Nb		0.0958	150	
^{95}Zr		0.175	150	
^{99}Tc		2.13×10^{5}	180	
^{106}Ru		1.02	810	39
^{129}I	Airborne	1.57×10^{7}		0.006
	Liquid		0.074	39
^{131}I	Airborne	0.022		0.0007
	Liquid			0.032
^{137}Cs	Airborne	30.0		0.002
	Liquid		4090	13
^{144}Ce		0.781	100	
^{241}Pu		14.4	1800	
U			11000 (kg)	

†Releases in 1978 only; 1.8 GW_e produced from fuel reprocessed.
‡Average for 3 plants normalized to 1985-1989; La Hague, France, Sellafield, UK, and Toka-Mura, Japan.
(From UNSCEAR 1993.)

non-gaseous activities are contained within the plant, but small releases occur to the environment according to limits set by the regulating authorities. These releases were rather large in earlier days, but have been considerably reduced as improved procedures and equipment are coming into operation, see Table 21.10. The authorized releases in the UK in 1978 were 10^4 TBq β-emitters (excl. T) per year. The ^{137}Cs release produced ~1000 Bq/l water in the middle of the Irish Sea (~150 km SW Windscale), and could be traced as far away as in the Baltic Sea. The release into the Atlantic in 1978 was ≤10 000 TBq. Not counting releases from plants in the US, former USSR and India, it is assumed to be <1000 TBq in 1992. It should be noted that the release of radionuclides into the sea from nuclear power operations is dominated by the effluents from a few reprocessing plants. It is, however, only a small fraction of the natural radioactivity in the oceans.

21.9. TREATMENT AND DEPOSITION OF LOW AND MEDIUM LEVEL WASTES

It has become a common practice at reprocessing plants to store low level solid waste (and sometimes higher level wastes) in trenches dug from the soil. These trenches, which commonly are 5–8 m deep, are sometimes lined with concrete or simply have a gravel bottom. For this purpose dry areas are selected (deserts, when available) or isolated areas with controlled groundwater conditions with respect to water table depth, flow rate, and direction. The disposed material normally should not exceed a dose rate of 1 mGy h^{-1} at 0.3 m distance, or contain a specific activity exceeding 1 MBq kg^{-1}. However, this varies, and may be a factor of 10 higher or lower in some places. When the trench is full, it is backfilled with earth, after which the surface dose rate usually is <0.01 mGy h^{-1}. Trenches of this kind are used in the USA, the UK, France, Russia, etc., where tens of thousands of cubic meters have been disposed annually; Fig. 21.17(A).

Since some of the waste products are longlived, and the physical protection of the waste in surface trenches is poor, radioactivity may ultimately leak into the groundwater, see Ch. 22. Therefore, in many countries repositories for final deposition of solid intermediate and low level wastes have been or are being built. The Swedish repository at Forsmark for ILW and LLW from reactors, hospitals, research facilities, etc. has been in operation since 1988; Fig. 21.17(B). It is located in solid granite rock below 50 m of sea water at the Baltic Sea coast. This underground site contains large concrete silos for the ILW and storage tunnels for ILW and LLW containers. The space between the concrete walls and the rock is filled with clay (bentonite). The repository is partly operated by remote control. Fig. 21.17(C) shows Belgian plans for a mixed ILW and HLW repository located in a large clay deposit. Also in this repository, concrete and clay are the main protective barriers. Geologic barriers are discussed in §21.13.